

CALCULUS

&

LINEAR ALGEBRA

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MODULE - 5

Linear Algebra in this Module begins with the recapitulation of matrix theory. **Rank of a matrix** by elementary transformations is discussed, followed by **Consistency** of a system of linear algebraic equations which can be deemed as an application of the concept of the rank of a matrix.

Exact solution of a system of linear algebraic equations by **Gauss and Gauss-Jordan elimination methods** are presented. Further **Gauss-Seidel method**, a numerical iterative method giving solution to the desired accuracy is discussed.

Definition of **Eigen values and Eigen vectors** of a square matrix along with the method of computation of all of them theoretically is presented. Further **Rayleigh's power method**, a numerical iterative method for the computation of the largest eigen value and the corresponding eigen vector is also discussed.

The module is concluded with the discussion of **Diagonalisation** of a square matrix of order two.

LINEAR ALGEBRA

5.1 Recapitulation of basic matrix theory

A set of mn elements written in an array of m rows and n columns embedded in brackets $[]$ or $()$ is called a **matrix** of order $m \times n$ (m by n).

The following array is a typical $m \times n$ matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} = [a_{ij}] \quad \uparrow$$

Order of $A = O(A) = m \times n$.

If $m = n$ the matrix is called a **square matrix** of order n .

The matrix of order $1 \times n$ is called a **row matrix** and a matrix of order $n \times 1$ is called a **column matrix**.

Examples : $[1 \ 2 \ 3] \dots$ Row matrix (order 1×3)

$$\begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} \dots \text{Column matrix (order } 3 \times 1 \text{)}$$

Consider a square matrix A of order n as follows.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

In A , the elements $a_{11}, a_{22}, \dots, a_{nn}$ constitute *principal diagonal* of the square matrix A . The sum of all these elements is called the *trace* of A . A square matrix having all the elements below the principal diagonal zero is called an **upper triangular matrix** and a square matrix having all the elements above the principal diagonal zero is called a **lower triangular matrix**.

Examples : $\begin{bmatrix} 1 & 2 & 5 \\ 0 & 4 & 7 \\ 0 & 0 & 8 \end{bmatrix}, \begin{bmatrix} 3 & 0 & 0 \\ 4 & 5 & 0 \\ 6 & 7 & 8 \end{bmatrix}$

are respectively upper and lower triangular matrices.

A square matrix is said to be a **diagonal matrix** if every element other than the principal diagonal elements are zero.

Examples : $\begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix}, \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

If every element of a diagonal matrix is the same then it is called a **Scalar matrix**.

Examples : $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

A scalar matrix in which each diagonal element is equal to unity, that is 1, is called a **Unit matrix** or **Identity matrix** usually denoted by I .

Examples : $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

A matrix having all its elements equal to zero is called a **null matrix**.

Examples : $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

The matrix obtained by interchanging their rows and columns is called as the **transpose** of the given matrix usually denoted by A' where A is the given matrix.

Obviously, $(A')' = A$.

Examples : If $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 4 \end{bmatrix}$ then $A' = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ -1 & 4 \end{bmatrix}$

If $A = \begin{bmatrix} 2 & -1 & 4 \end{bmatrix}$ then $A' = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$

A square matrix A is said to be *symmetric* if $A = A'$ and *skew symmetric* if $A = -A'$.

Examples : $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$, $\begin{bmatrix} 1 & 4 & 2 \\ 4 & 5 & 0 \\ 2 & 0 & 7 \end{bmatrix}$ are symmetric matrices.

(observe that row elements and column elements are the same)

$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & 4 \\ 1 & -4 & 0 \end{bmatrix}$ are skew symmetric matrices.

(Observe that the principal diagonal elements are zero and column elements are negative of the row elements.)

Algebra of matrices

If A is any matrix and k is any scalar, the matrix obtained by multiplying every element of A by k is called the scalar multiple of the matrix A denoted by kA .

Examples :

If $A = \begin{bmatrix} 2 & 3 & 5 \\ -1 & 4 & 6 \end{bmatrix}$ then $3A = \begin{bmatrix} 6 & 9 & 15 \\ -3 & 12 & 18 \end{bmatrix}$, $\frac{1}{2} A = \begin{bmatrix} 1 & 3/2 & 5/2 \\ -1/2 & 2 & 3 \end{bmatrix}$

Sum and Difference of two matrices

If A and B are two matrices of the same order, their sum $A + B$, difference $A - B$ is the sum/difference of the corresponding elements.

Examples

If $A = \begin{bmatrix} -1 & 4 \\ 5 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 6 & 3 \\ 5 & -4 \end{bmatrix}$ then

$$A + B = \begin{bmatrix} 5 & 7 \\ 10 & 0 \end{bmatrix}, A - B = \begin{bmatrix} -7 & 1 \\ 0 & 8 \end{bmatrix}$$

$$2A + 3B = \begin{bmatrix} 16 & 17 \\ 25 & -4 \end{bmatrix}, A + \frac{1}{2} B = \begin{bmatrix} 2 & 11/2 \\ 15/2 & 2 \end{bmatrix}$$

Product of two matrices

If A is a matrix of order $m \times n$ and B is a matrix of order $n \times p$ then the product AB exists and will be a matrix of order $m \times p$.

Row elements of A are multiplied with the corresponding column elements of B and are added.

Examples :

1. Consider, $A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & 4 \\ 3 & 6 \end{bmatrix}$

Order of A , $O(A) = 2 \times 3$, $O(B) = 3 \times 2 \therefore O(AB) = 2 \times 2$

The product AB is computed as follows.

$$AB = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 4 \\ 3 & 6 \end{bmatrix}$$

$$AB = \begin{bmatrix} 3 \times 1 + 2 \times 0 + 1 \times 3, & 3 \times 2 + 2 \times 4 + 1 \times 6 \\ 2 \times 1 + 4 \times 0 + 2 \times 3, & 2 \times 2 + 4 \times 4 + 2 \times 6 \end{bmatrix} = \begin{bmatrix} 6 & 20 \\ 8 & 32 \end{bmatrix}$$

Also let us examine the possibility of the computation of the matrix product BA .

$$O(B) = 3 \times 2, O(A) = 2 \times 3 \therefore O(BA) = 3 \times 3$$

The product BA is computed as follows.

$$BA = \begin{bmatrix} 1 & 2 \\ 0 & 4 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 \times 3 + 2 \times 2, & 1 \times 2 + 2 \times 4, & 1 \times 1 + 2 \times 2 \\ 0 \times 3 + 4 \times 2, & 0 \times 2 + 4 \times 4, & 0 \times 1 + 4 \times 2 \\ 3 \times 3 + 6 \times 2, & 3 \times 2 + 6 \times 4, & 3 \times 1 + 6 \times 2 \end{bmatrix}$$

$$\text{ie., } BA = \begin{bmatrix} 7 & 10 & 5 \\ 8 & 16 & 8 \\ 21 & 30 & 15 \end{bmatrix}$$

Remark : We come across with product of two square matrices of the same order. Obviously the resulting product matrix will also be a square matrix of the same order.

Note : Properties of matrix multiplication

1. The matrix product AB may exist but BA may not exist. Even if both exist, $AB \neq BA$. Matrix multiplication in general is not commutative.
2. If $O(A) = m \times n$, $O(B) = n \times p$, $O(C) = p \times q$ then $A(BC) = (AB)C$ which implies that the matrix multiplication is associative.
3. If $O(A) = m \times n$ and $O(B) = n \times p = O(C)$ then $A(B+C) = AB + AC$, which implies that matrix multiplication is distributive.
4. $(AB)' = B'A'$
5. If A is any square matrix, the matrix product $A \cdot A$ is denoted by A^2 . Further $A \cdot A^2 = A^3$ and so-on.
In general if p and q are positive integers,
 $A^p \cdot A^q = A^{p+q}$ and $(A^p)^q = A^{pq}$.

Orthogonal Matrix

A square matrix A is said to be *orthogonal* if $AA' = I = A' A$

Singular matrix and Nonsingular matrix

A square matrix A is said to be *singular* if its determinant is zero and A is said to be *nonsingular* if its determinant is not equal to zero.

That is, $|A| = 0 \Rightarrow A$ is singular.

$|A| \neq 0 \Rightarrow A$ is nonsingular.

Inverse of a square matrix

If A and B are two square matrices of the same order such that

$$AB = BA = I$$

then B is called the inverse of A denoted by A^{-1} .

The necessary and the sufficient condition for a square matrix A to possess inverse is that $|A| \neq 0$. That is to say that the inverse of a square matrix exists, if and only if the matrix A is nonsingular.

Note : 1. The inverse of a square matrix is unique.

$$2. (AB)^{-1} = B^{-1} A^{-1}$$

3. We have already said that $AA' = I = A' A \Rightarrow A$ is orthogonal.

Comparing this with the definition of the inverse of a square matrix we can also conclude that

$$A' = A^{-1} \Rightarrow A \text{ is orthogonal.}$$

Adjoint of a square matrix

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Then the adjoint of A denoted by $\text{Adj } A$ is as follows.

$$\text{Adj } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}' = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

Here A_{ij} represents the cofactor of a_{ij} in $|A|$. That is, $A_{ij} = (-1)^{i+j}$ multiplied by the value of the determinant obtained by deleting the i^{th} row and j^{th} column. However in the case of a second order square matrix, the cofactors will be a single element.

If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then $\text{Adj } A = \begin{bmatrix} +a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$

Now, for the third order square matrix A we have,

$$\text{Adj } A = \begin{bmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, & - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}, & + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}, & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \\ + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}, & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}, & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}$$

Remark : It is preferable to write the cofactors of elements column wise premultiplied by the signs $+, -, +, -, +, -, + \dots$ and enter row wise in the matrix, resulting in the adjoint of the given matrix.

Further the adjoint of a square matrix helps in finding the inverse of a nonsingular matrix A by the following established result.

$$A^{-1} = \frac{1}{|A|} \text{Adj } A$$

Illustrative examples to compute the inverse of a given square matrix.

$$[1] \quad A = \begin{bmatrix} 1 & -2 \\ -3 & 8 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & -2 \\ -3 & 8 \end{vmatrix} \\ = 8 - (+6) = 2$$

$$\text{Adj } A = \begin{bmatrix} +8, & -(-2) \\ -(-3), & 1 \end{bmatrix} = \begin{bmatrix} 8 & 2 \\ 3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 8 & 2 \\ 3 & 1 \end{bmatrix}$$

$$[2] \quad A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & -6 & -7 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & -6 & -7 \end{vmatrix} \\ = 1(-28 + 30) - 0 - 1(-18 - 0) = 20$$

$$\text{Adj } A = \begin{bmatrix} +(-28 + 30), & -(0 - 6), & +(0 + 4) \\ -(-21 - 0), & +(-7 - 0), & -(5 + 3) \\ +(-18 - 0), & -(-6 - 0), & +(4 - 0) \end{bmatrix} = \begin{bmatrix} 2 & 6 & 4 \\ 21 & -7 & -8 \\ -18 & 6 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{20} \begin{bmatrix} 2 & 6 & 4 \\ 21 & -7 & -8 \\ -18 & 6 & 4 \end{bmatrix}$$

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5.2 Rank of a matrix by elementary transformations

• *Elementary transformations (operations) associated with a matrix.*

The following are the elementary *row transformations* of a matrix. The transformations can also be applied for columns.

- 1. Interchange of any two rows.
- 2. Multiplication of any row by a non zero constant.
- 3. Addition to any row a constant multiple of any other row.

Elementary transformations along with the notations we use, is illustrated in the following table by considering the matrix.

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

Example

	Elementary row transformation	Notation	Resultant of a matrix A
1.	Interchange of first and second row	$R_1 \leftrightarrow R_2$	$\begin{bmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$
2.	Multiplication of third row by a constant k	$k R_3$	$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ kc_1 & kc_2 & kc_3 \end{bmatrix}$
3.	Addition to second row k times the first row	$R_2 \rightarrow kR_1 + R_2$	$\begin{bmatrix} a_1 & a_2 & a_3 \\ (ka_1 + b_1) & (ka_2 + b_2) & (ka_3 + b_3) \\ c_1 & c_2 & c_3 \end{bmatrix}$

Equivalent matrices : Two matrices A and B of the same order are said to be *equivalent* if one matrix can be obtained from the other by a finite number of successive elementary row (column) transformations. We use the notation $A \sim B$.

• Echelon form of a matrix

A non zero matrix A is said to be in *row echelon form* if the following conditions prevail.

- All the zero rows are below non zero rows.
- The first non zero entry in any non zero row is 1.
(Obviously in view of the condition (a) the entries below 1 in the same column are zero.)

Examples :

$$1. \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$2. \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$3. \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

• Rank of a matrix

The *rank* of a matrix A in its **echelon form** is equal to the *number of non zero rows*. It is denoted by $\rho (A)$.

Flow chart for solving problems - Finding the rank of a matrix

In order to reduce the given matrix to a row echelon form we must prefer to have the leading entry (*first entry in the first row*) non zero, much preferably 1.



In case this entry is zero we can interchange with any suitable row to meet the requirement.



We then focus on the leading non zero entry (starting from the first row) to make all the elements in that column zero. However the transformation has to be performed for the entire row.



Row echelon form will be achieved first and we can instantly write down the rank, which being the number of zero rows.

Note : It is advisable to avoid fractions as far as possible during the process of elementary transformations.

Further we can say that if r is the rank of a matrix A of order $m \times n$ ($r \leq m$), r number of rows of the matrix are linearly independent.

Observe the following illustrative examples.

$$[1] \quad A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

Performing, $R_2 \rightarrow -3R_1 + R_2$

$$A \sim \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \Rightarrow \boxed{\rho(A) = 1}$$

This means that *one row* of the matrix A is *linearly independent*. It can be easily seen that $R_2 = 3R_1$ or $R_1 = 1/3 \cdot R_2$

$$[2] \quad A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 2 & 3 \\ 1 & 5 & 7 \end{bmatrix}$$

$R_1 \leftrightarrow R_2$ yields,

$$A \sim \begin{bmatrix} -1 & 2 & 3 \\ 2 & 3 & 4 \\ 1 & 5 & 7 \end{bmatrix}$$

$R_2 \rightarrow 2R_1 + R_2$, $R_3 \rightarrow R_1 + R_3$ gives us

$$A \sim \begin{bmatrix} -1 & 2 & 3 \\ 0 & 7 & 10 \\ 0 & 7 & 10 \end{bmatrix}$$

$R_3 \rightarrow -R_2 + R_3$ gives,

$$A \sim \begin{bmatrix} -1 & 2 & 3 \\ 0 & 7 & 10 \\ 0 & 0 & 0 \end{bmatrix}$$

Finally $-R_1$ and $1/7 \cdot R_2$ yields

$$A \sim \begin{bmatrix} 1 & -2 & -3 \\ 0 & 1 & 10/7 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \boxed{\rho(A) = 2}$$

This means that *two rows* of the matrix A are linearly independent. It can be easily seen that $R_1 + R_2 = R_3$ or $R_2 = R_3 - R_1$ or $R_1 = R_3 - R_2$.

Any one row of the matrix A is expressible in terms of the other two rows.

WORKED PROBLEMS

Find the rank of the following matrices by elementary row transformations.

$$[1] \quad A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

$$R_2 \rightarrow -2R_1 + R_2, \quad R_3 \rightarrow -R_1 + R_3$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 1 & 3 \end{bmatrix}$$

$$R_3 \rightarrow R_2 + R_3$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow (-1)R_2$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrix A in the row echelon form is having *two* non zero rows.

Thus,

$$\boxed{\rho(A) = 2}$$

$$[2] \quad A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$$

$$\Rightarrow R_1 \leftrightarrow R_2$$

$$A \sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$$

$$R_3 \rightarrow -2R_1 + R_3$$

$$A \sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \end{bmatrix}$$

$$R_3 \rightarrow -R_2 + R_3$$

$$A \sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$1/2 \cdot R_1, 1/2 \cdot R_2$$

$$A \sim \begin{bmatrix} 1 & 3/2 & 5/2 & 2 \\ 0 & 1 & 3/2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrix A in the row echelon form is having *two* non zero rows.

Thus,

$$\boxed{\rho(A) = 2}$$

$$[3] \quad A = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$\Rightarrow R_1 \leftrightarrow R_2$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_2 \rightarrow -2R_1 + R_2, R_3 \rightarrow -R_1 + R_3$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & -5 & -9 & 1 \\ 0 & -2 & -2 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_2 \leftrightarrow R_4$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -2 & -2 & 2 \\ 0 & -5 & -9 & 1 \end{bmatrix}$$

$$R_3 \rightarrow 2R_2 + R_3, R_4 \rightarrow 5R_2 + R_4$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -4 & -4 \end{bmatrix}$$

$$R_3 \leftrightarrow R_4 \text{ and } -1/4 \cdot R_3$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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The matrix A in the row echelon form is having *three* non zero rows.

Thus,

$$\rho(A) = 3$$

$$[4] \quad A = \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$$

[Dec 2017]

☞ [Firstly we prefer to interchange the first and second rows as it would be convenient to make the leading entry in the other rows zero]

$$R_2 \leftrightarrow R_1$$

$$A \sim \begin{bmatrix} 2 & 1 & 3 & 4 \\ 4 & 0 & 2 & 1 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$$

$$R_2 \rightarrow -2R_1 + R_2, R_3 \rightarrow -R_1 + R_3, R_4 \rightarrow -R_1 + R_4$$

$$A \sim \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & -2 & -4 & -7 \\ 0 & 2 & 1 & 3 \\ 0 & 2 & -2 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_2 + R_3, R_4 \rightarrow R_2 + R_4$$

$$A \sim \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & -2 & -4 & -7 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & -6 & -7 \end{bmatrix}$$

$$R_4 \rightarrow -2R_3 + R_4$$

$$A \sim \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & -2 & -4 & -7 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$1/2 \cdot R_1, -1/2 \cdot R_2, -1/3 \cdot R_3$$

$$A \sim \begin{bmatrix} 1 & 1/2 & 3/2 & 2 \\ 0 & 1 & 2 & 7/2 \\ 0 & 0 & 1 & 4/3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

All the *four* rows are non zero in the row echelon form A .

Thus,

$$\boxed{\rho(A) = 4}$$

$$[5] \quad A = \begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ -2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow 2R_1 + R_2, R_3 \rightarrow -R_1 + R_3$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 3 & 3 & 1 \\ 0 & -2 & -2 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$$R_2 \leftrightarrow R_4$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & -2 & -2 & 0 \\ 0 & 3 & 3 & 1 \end{bmatrix}$$

$$R_3 \rightarrow 2R_2 + R_3, R_4 \rightarrow -3R_2 + R_4$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

$$R_4 \rightarrow R_3 + R_4$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrix A in the row echelon form has *three* non zero rows.

Thus,

$$\boxed{\rho(A) = 3}$$

$$[6] \quad A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

[June 2018]

$$\Rightarrow R_1 \leftrightarrow R_2$$

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

$$R_3 \rightarrow -3R_1 + R_3, R_4 \rightarrow -R_1 + R_4$$

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix}$$

$$R_3 \rightarrow -R_2 + R_3, R_4 \rightarrow -R_2 + R_4$$

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrix A in the row echelon form has *two* non zero rows.

Thus,

$$\rho(A) = 2$$

$$[7] \quad A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

[June 2016]

$$R_1 \leftrightarrow R_2$$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

[We can perform elementary transformations like the earlier problems. However if one can observe that the fourth row is the sum of the first three rows we can as well proceed by subtracting the fourth row from the sum of first three rows]

$$R_4 \rightarrow -(R_1 + R_2 + R_3) + R_4$$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow -2R_1 + R_2, \quad R_3 \rightarrow -3R_1 + R_3$$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrix A in the row echelon form has *three* non zero rows.

Thus,

$$\rho(A) = 3$$

ASSIGNMENT

Find the rank of the following matrices by elementary row transformations.

1. $\begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \end{bmatrix}$

2. $\begin{bmatrix} 1 & -3 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 3 & 4 & 1 & -2 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 3 & 4 & 5 \\ 3 & 2 & 5 & 2 \\ 2 & -1 & 1 & -3 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 3 & 5 & 4 \\ 2 & 1 & 5 \end{bmatrix}$

5. $\begin{bmatrix} 1 & -2 & -1 & 3 \\ 2 & 5 & -4 & 7 \\ -1 & -2 & -1 & 2 \\ 3 & 3 & -5 & 10 \end{bmatrix}$

6. $\begin{bmatrix} 8 & 2 & 1 & 6 \\ 2 & 1 & 0 & 1 \\ 3 & 0 & 1 & 3 \\ 5 & 1 & 1 & 4 \end{bmatrix}$

ANSWERS

1. 3

2. 3

3. 2

4. 2

5. 3

6. 3

5.3 Consistency of a system of linear equations

Consider a set of m linear equations in n unknowns as follows.

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \dots\dots\dots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{array} \right.$$

where a_{ij} 's and b_i 's are constants.

If $b_1, b_2, \dots, b_m$ are all zero, the system is said to be *homogeneous*.

The set of values $x_1, x_2, \dots, x_n$ which satisfy all the equations simultaneously is called a **solution** of the system of equations.

A system of linear equations is said to be **consistent** if it possess a solution. Otherwise the system is said to be **inconsistent**.

The above system of equations can be written in the matrix form,

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{bmatrix}$$

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That is, $AX = B$ where the matrix A is of order $m \times n$, X is of order $n \times 1$ and hence their product denoted by B is of order $m \times 1$. Obviously $AX = [0]$ is the matrix representation of the homogeneous set of equations where $[0]$ is a null matrix of order $m \times 1$.

$x_1 = x_2 = x_3 = \dots = x_n = 0$ is obviously a solution of the homogeneous system of equations and is called a **trivial solution**. If at least one x_i ($i = 1, 2, \dots, n$) is not equal to zero then it is called a **non trivial solution**.

The system of equations $AX = B$ may or may not possess a solution. If the system possesses a solution, it may or may not be the only solution. The concept of the rank of a matrix helps us to conclude (i) whether the system is consistent or not (ii) whether the system possess unique solution or many solutions.

● Condition for consistency and types of solution

Consider a system of m equations in n unknowns represented in the matrix form $AX = B$.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}; X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}; B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

A is called the *coefficient matrix*.

The matrix formed by appending to A an extra column consisting of the elements of B is called the *augmented matrix* denoted by $[A : B]$.

That is,
$$[A : B] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & : & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & : & b_2 \\ \dots & \dots & \dots & \dots & : & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & : & b_m \end{bmatrix}$$

The system of equations represented by the matrix equation $AX = B$ is *consistent* if $\rho[A] = \rho[A : B]$.

Suppose, $\rho[A] = \rho[A : B] = r$, then the condition for two types of solution are as follows.

1. **Unique solution** : $\rho[A] = \rho[A : B] = r = n$, n being the number of unknowns.
2. **Infinite solutions** : $\rho[A] = \rho[A : B] = r < n$.

In this case $(n - r)$ unknowns can take arbitrary values. Obviously $\rho[A] \neq \rho[A : B]$ implies that the system is *inconsistent* (does not possess a solution).

Working procedure for problems

- Step - 1.** We first form the augmented matrix $[A : B]$ and we can clearly identify the portion of the coefficient matrix A in it.
- Step - 2.** We reduce the matrix $[A : B]$ to an echelon form by elementary row transformations. This will enable us to immediately write down

the rank of A and also $[A : B]$, with the result we can decide the consistency aspect of the system of equations.

Step - 3. The echelon form of $[A : B]$ is converted back to the equation form and the solution will emerge easily.

WORKED PROBLEMS

[8] Test for consistency and solve :

$$x + y + z = 6$$

$$x - y + 2z = 5$$

$$3x + y + z = 8$$

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & -1 & 2 & : & 5 \\ 3 & 1 & 1 & : & 8 \end{bmatrix} \text{ is the augmented matrix.}$$

We now perform elementary row operations,

$$R_2 \rightarrow -R_1 + R_2, R_3 \rightarrow -3R_1 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & -2 & 1 & : & -1 \\ 0 & -2 & -2 & : & -10 \end{bmatrix}$$

$$R_3 \rightarrow -R_2 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & -2 & 1 & : & -1 \\ 0 & 0 & -3 & : & -9 \end{bmatrix}$$

Note : We need not make the leading non zero entry in every row 1 as we can decide on the rank of the matrices A and $[A : B]$ at this stage.

Both A and $[A : B]$ matrices have all the three rows non zero.

$\therefore \rho[A] = 3$ and $\rho[A : B] = 3$. That is, $r = 3$.

Also, the number of independent variables $n = 3$.

Since $\rho[A] = \rho[A : B] = 3$ ($r = n = 3$) the given system of equations is consistent and will have unique solution.

Let us now convert the prevailing form of $[A : B]$ into a set of equations as follows.

$$x + y + z = 6 \quad \dots (i)$$

$$-2y + z = -1 \quad \dots (ii)$$

$$-3z = -9 \quad \dots (iii)$$

From (iii) $z = 3$, substituting this value in (ii) we get $y = 2$. Finally substituting these values in (i) we get $x = 1$.

Thus, $\boxed{x = 1, y = 2, z = 3}$ is the unique solution.

[9] Test for consistency and solve

$$x + 2y + 3z = 14$$

$$4x + 5y + 7z = 35$$

$$3x + 3y + 4z = 21$$

$$\Rightarrow [A : B] = \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 4 & 5 & 7 & : & 35 \\ 3 & 3 & 4 & : & 21 \end{bmatrix} \text{ is the augmented matrix.}$$

$$R_2 \rightarrow -4R_1 + R_2, \quad R_3 \rightarrow -3R_1 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 0 & -3 & -5 & : & -21 \\ 0 & -3 & -5 & : & -21 \end{bmatrix}$$

$$R_3 \rightarrow -R_2 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 0 & -3 & -5 & : & -21 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

We have $\rho[A] = 2$, $\rho[A : B] = 2$. That is, $r = 2$. Also $n = 3$.

Since $\rho[A] = \rho[A : B] = 2 < 3$, ($r < n$) **the system is consistent and will have infinite solutions.** Here $(n - r) = 1$ and hence one of the variables can take arbitrary values.

We now have, $x + 2y + 3z = 14$... (i)

$-3y - 5z = -21$... (ii)

Let, $z = k$ be arbitrary.

$\therefore$ from (ii) $-3y - 5k = -21$ or $y = \frac{21 - 5k}{3} = 7 - \frac{5k}{3}$

Now, from (i), $x + 2\{7 - (5k/3)\} + 3k = 14 \therefore x = k/3$

Thus, $\boxed{x = \frac{k}{3}, y = 7 - \frac{5k}{3}, z = k}$ represent infinite solutions, since k is arbitrary.

[10] Test for consistency and solve :

$$x - 4y + 7z = 14$$

$$3x + 8y - 2z = 13$$

$$7x - 8y + 26z = 5$$

$[A : B] = \begin{bmatrix} 1 & -4 & 7 & : & 14 \\ 3 & 8 & -2 & : & 13 \\ 7 & -8 & 26 & : & 5 \end{bmatrix}$ is the augmented matrix.

$$R_2 \rightarrow -3R_1 + R_2, R_3 \rightarrow -7R_1 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & -4 & 7 & : & 14 \\ 0 & 20 & -23 & : & -29 \\ 0 & 20 & -23 & : & -93 \end{bmatrix}$$

$$R_3 \rightarrow -R_2 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & -4 & 7 & : & 14 \\ 0 & 20 & -23 & : & -29 \\ 0 & 0 & 0 & : & -64 \end{bmatrix}$$

We have, $\rho[A] = 2$ and $\rho[A : B] = 3$

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Since $\rho[A] \neq \rho[A:B]$ the system is **inconsistent**. (does not possess any solution)

Remark : By the conversion of the above form of the matrix to equations, the last row will result in an identity of the form $0 = -64$ which is absurd. Hence no solution for the system.

[11] Test for consistency and solve :

$$5x_1 + x_2 + 3x_3 = 20$$

$$2x_1 + 5x_2 + 2x_3 = 18$$

$$3x_1 + 2x_2 + x_3 = 14$$

$$[A:B] = \begin{bmatrix} 5 & 1 & 3 & : & 20 \\ 2 & 5 & 2 & : & 18 \\ 3 & 2 & 1 & : & 14 \end{bmatrix} \text{ is the augmented matrix.}$$

$$R_2 \rightarrow -2R_1 + 5R_2, \quad R_3 \rightarrow -3R_1 + 5R_3$$

$$[A:B] \sim \begin{bmatrix} 5 & 1 & 3 & : & 20 \\ 0 & 23 & 4 & : & 50 \\ 0 & 7 & -4 & : & 10 \end{bmatrix}$$

$$R_3 \rightarrow -7R_2 + 23R_3$$

$$[A:B] \sim \begin{bmatrix} 5 & 1 & 3 & : & 20 \\ 0 & 23 & 4 & : & 50 \\ 0 & 0 & -120 & : & -120 \end{bmatrix}$$

We have $\rho[A] = 3$, $\rho[A:B] = 3$. That is, $r = 3$. Also $n = 3$.

Since $\rho[A] = \rho[A:B] = 3$ ($r = n = 3$) the system is consistent and will have unique solution.

$$\text{We now have,} \quad 5x_1 + x_2 + 3x_3 = 20 \quad \dots (i)$$

$$23x_2 + 4x_3 = 50 \quad \dots (ii)$$

$$-120x_3 = -120 \quad \dots (iii)$$

$\therefore$ from (iii), $x_3 = 1$.

From (ii) we get $x_2 = 2$ and from (i) we get $x_1 = 3$

Thus $\boxed{x_1 = 3, x_2 = 2, x_3 = 1}$ is the unique solution.

[12] Test for consistency and solve :

$$x + 2y + 2z = 1$$

$$2x + y + z = 2$$

$$3x + 2y + 2z = 3$$

$$y + z = 0$$

$$[A : B] = \begin{bmatrix} 1 & 2 & 2 & : & 1 \\ 2 & 1 & 1 & : & 2 \\ 3 & 2 & 2 & : & 3 \\ 0 & 1 & 1 & : & 0 \end{bmatrix} \text{ is the augmented matrix}$$

$$R_2 \rightarrow -2R_1 + R_2, R_3 \rightarrow -3R_1 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 2 & 2 & : & 1 \\ 0 & -3 & -3 & : & 0 \\ 0 & -4 & -4 & : & 0 \\ 0 & 1 & 1 & : & 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_4$$

$$[A : B] \sim \begin{bmatrix} 1 & 2 & 2 & : & 1 \\ 0 & 1 & 1 & : & 0 \\ 0 & -4 & -4 & : & 0 \\ 0 & -3 & -3 & : & 0 \end{bmatrix}$$

$$R_3 \rightarrow 4R_2 + R_3, R_4 \rightarrow 3R_2 + R_4$$

$$[A : B] \sim \begin{bmatrix} 1 & 2 & 2 & : & 1 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

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We have $\rho[A] = 2$, $\rho[A : B] = 2$. That is, $r = 2$. Also $n = 3$.

Since $\rho[A] = \rho[A : B] = 2 < 3$ ($r < n$) the system is consistent and will have infinite solutions.

We now have, $x + 2y + 2z = 1$... (i)

$$y + z = 0 \quad \dots (ii)$$

Let $z = k$ be arbitrary. $\therefore y = -k$ from (ii).

Also from (i), $x + (-2k) + 2k = 1 \therefore x = 1$

Thus, $x = 1, y = -k, z = k$ are the infinite number of solutions since k is arbitrary.

[13] Show that the following system of equations does not possess any solution.

$$5x + 3y + 7z = 5$$

$$3x + 26y + 2z = 9$$

$$7x + 2y + 10z = 5$$

$[A : B] = \begin{bmatrix} 5 & 3 & 7 & : & 5 \\ 3 & 26 & 2 & : & 9 \\ 7 & 2 & 10 & : & 5 \end{bmatrix}$ is the augmented matrix.

$$R_2 \rightarrow -3R_1 + 5R_2, R_3 \rightarrow -7R_1 + 5R_3$$

$$[A : B] \sim \begin{bmatrix} 5 & 3 & 7 & : & 5 \\ 0 & 121 & -11 & : & 30 \\ 0 & -11 & 1 & : & -10 \end{bmatrix}$$

$$R_3 \rightarrow R_2 + 11R_3$$

$$[A : B] \sim \begin{bmatrix} 5 & 3 & 7 & : & 5 \\ 0 & 121 & -11 & : & 30 \\ 0 & 0 & 0 & : & -80 \end{bmatrix}$$

We have, $\rho[A] = 2, \rho[A : B] = 3$.

Since $\rho[A] \neq \rho[A : B]$ the system is inconsistent.

[14] Investigate the values of λ and μ such that the system of equations

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu, \text{ may have}$$

(a) Unique solution (b) Infinite solution (c) No solution.

$$\mathcal{E} \quad [A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix} \text{ is the augmented matrix.}$$

$$R_2 \rightarrow -R_1 + R_2, R_3 \rightarrow -R_1 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 1 & \lambda - 1 & : & \mu - 6 \end{bmatrix}$$

$$R_3 \rightarrow -R_2 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & \lambda - 3 & : & \mu - 10 \end{bmatrix}$$

- (a) *Unique solution* : We must have $\rho[A] = \rho[A : B] = 3$, $\rho[A]$ will be 3 if $(\lambda - 3) \neq 0$ since the other two entries in the last row of A are zero. If $(\lambda - 3) \neq 0$ or $\lambda \neq 3$ irrespective of the value of μ , $\rho[A : B]$ will also be 3.

Hence, the system will have **unique solution** if $\lambda \neq 3$.

- (b) *Infinite solutions* : Here we have $n = 3$ and we need

$\rho[A] = \rho[A : B] = r < 3$. We must have $r = 2$ since first row and second row are non zero.

$\therefore \rho[A] = \rho[A : B] = 2$ only when the last row of $[A : B]$ is completely zero. This is possible if $\lambda - 3 = 0$, $\mu - 10 = 0$

Hence, the system will have **infinite solution** if $\lambda = 3$ and $\mu = 10$.

- (c) *No solution* : We must have $\rho[A] \neq \rho[A : B]$. By case (a) $\rho[A] = 3$ if $\lambda \neq 3$ and hence if $\lambda = 3$ we obtain $\rho[A] = 2$.

If we impose $(\mu - 10) \neq 0$ then $\rho[A : B]$ will be 3.

Hence, the system has **no solution** if $\lambda = 3$ and $\mu \neq 10$.

[15] Find for what values of k the system of equations

$$x + y + z = 1$$

$$x + 2y + 4z = k$$

$$x + 4y + 10z = k^2$$

possesses a solution. Solve completely in each case.

$$\Rightarrow [A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 1 & 2 & 4 & : & k \\ 1 & 4 & 10 & : & k^2 \end{bmatrix} \text{ is the augmented matrix.}$$

$$R_2 \rightarrow -R_1 + R_2, \quad R_3 \rightarrow -R_1 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & 1 & 3 & : & k-1 \\ 0 & 3 & 9 & : & k^2-1 \end{bmatrix}$$

$$R_3 \rightarrow -3R_2 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & 1 & 3 & : & k-1 \\ 0 & 0 & 0 & : & k^2-3k+2 \end{bmatrix}$$

$\rho[A] = 2$ and for the system to be consistent we must have $\rho[A : B]$

also 2. This is possible if $k^2 - 3k + 2 = 0 \therefore k = 1, k = 2$.

Hence we conclude that the system possesses a solution if $k = 1, 2$,

Since $\rho[A] = \rho[A : B] = 2 < 3$ for the cases $k = 1, 2$ the system will have infinite solution and the same are as follows.

Case - (i) : $k = 1$. The system of equations are,

$$x + y + z = 1 \quad \dots (i)$$

$$y + 3z = 0 \quad \dots (ii)$$

Let $z = k_1$ be arbitrary $\therefore$ from (ii) $y = -3k_1$ and from (i) $x = 1 + 2k_1$

Case - (ii) : $k = 2$. The system of equations are,

$$x + y + z = 1 \quad \dots (iii)$$

$$y + 3z = 1 \quad \dots (iv)$$

Let $z = k_2$ be arbitrary $\therefore$ from (iv) $y = 1 - 3k_2$ and from (iii) $x = 2k_2$

Thus, $x = 1 + 2k_1, y = -3k_1, z = k_1$ and $x = 2k_2, y = 1 - 3k_2, z = k_2$

5.4 Solution of a system of linear equations

5.41 Gauss elimination method

This method is illustrated by considering a system of three independent equations in three unknowns.

Consider the system of equations

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \right\} \dots (i)$$

The system (i) is equivalent to the matrix equation,

$$AX = B \dots (ii)$$

where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

The method aims in reducing the coefficient matrix A to an upper triangular matrix.

We consider the augmented matrix $[A : B]$ where

$$[A : B] = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right]$$

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Step - 1. We use the element a_{11} ($\neq 0$) to make the elements a_{21} and a_{31} zero by elementary row transformations.

This transforms $[A : B]$ into the form

$$[A : B] \sim \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & a'_{32} & a'_{33} & b'_3 \end{array} \right] \dots (iii)$$

Step - 2. We use the element a'_{22} ($\neq 0$) to make the element a'_{32} zero by elementary row transformation.

This transforms $[A : B]$ as in (iii) into the form

$$[A : B] \sim \begin{bmatrix} a_{11} & a_{12} & a_{13} & : & b_1 \\ 0 & a_{22}' & a_{23}' & : & b_2' \\ 0 & 0 & a_{33}'' & : & b_3'' \end{bmatrix} \dots (iv)$$

(Observe that the matrix A is reduced to the upper triangular form)

From (iv), the given system of linear equations is equivalent to the system of equations

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{22}'x_2 + a_{23}'x_3 = b_2'$$

$$a_{33}''x_3 = b_3''$$

We get x_3 from the last of the equation and by back substitution we get x_2 and x_1 . The values of x_1, x_2, x_3 so obtained constitutes the exact solution of the given system of equations.

Note : In case a_{11} or a_{22}' is zero we only need to re-arrange the equations.

WORKED PROBLEMS

[16] Solve the following system of equations by Gauss elimination method.

[Dec 2017]

$$x + y + z = 9$$

$$x - 2y + 3z = 8$$

$$2x + y - z = 3$$

☞ The augmented matrix of the system is,

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 1 & -2 & 3 & : & 8 \\ 2 & 1 & -1 & : & 3 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{matrix}$$

$$R_2 \rightarrow -R_1 + R_2, R_3 \rightarrow -2R_1 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -3 & 2 & : & -1 \\ 0 & -1 & -3 & : & -15 \end{bmatrix}$$

$$R_3 \rightarrow R_2 + (-3)R_1$$

$$R_3 \rightarrow R_2 + (-3)R_1$$

$$[A:B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -3 & 2 & : & -1 \\ 0 & 0 & 11 & : & 44 \end{bmatrix}$$

Hence we have, $x + y + z = 9$
 $-3y + 2z = -1$
 $11z = 44 \therefore z = 4$

By back substitution, $-3y + 8 = -1 \therefore y = 3$; Also $x = 2$.

Thus $x = 2, y = 3, z = 4$ is the required solution.

[17] Solve by Gauss elimination method :

[June 2018]

$$2x + y + 4z = 12$$

$$4x + 11y - z = 33$$

$$8x - 3y + 2z = 20$$

☞ The augmented matrix of the system is,

$$[A:B] = \begin{bmatrix} 2 & 1 & 4 & : & 12 \\ 4 & 11 & -1 & : & 33 \\ 8 & -3 & 2 & : & 20 \end{bmatrix}$$

$$R_2 \rightarrow -2R_1 + R_2, R_3 \rightarrow -4R_1 + R_3$$

$$[A:B] \sim \begin{bmatrix} 2 & 1 & 4 & : & 12 \\ 0 & 9 & -9 & : & 9 \\ 0 & -7 & -14 & : & -28 \end{bmatrix}$$

$$1/9 \cdot R_2, -1/7 \cdot R_3$$

$$[A:B] \sim \begin{bmatrix} 2 & 1 & 4 & : & 12 \\ 0 & 1 & -1 & : & 1 \\ 0 & 1 & 2 & : & 4 \end{bmatrix}$$

$$R_3 \rightarrow -R_2 + R_3$$

$$[A:B] \sim \begin{bmatrix} 2 & 1 & 4 & : & 12 \\ 0 & 1 & -1 & : & 1 \\ 0 & 0 & 3 & : & 3 \end{bmatrix}$$

Hence we have, $2x + y + 4z = 12$
 $y - z = 1$
 $3z = 3 \therefore z = 1$

By back substitution, $y = 2$ and $x = 3$.

Thus, $x = 3, y = 2, z = 1$ is the required solution.

[18] Solve by Gauss elimination method :

$$x_1 - 2x_2 + 3x_3 = 2$$

$$3x_1 - x_2 + 4x_3 = 4$$

$$2x_1 + x_2 - 2x_3 = 5$$

☞ The augmented matrix associated with the given system of equations is,

$$[A:B] = \begin{bmatrix} 1 & -2 & 3 & : & 2 \\ 3 & -1 & 4 & : & 4 \\ 2 & 1 & -2 & : & 5 \end{bmatrix}$$

$$R_2 \rightarrow -3R_1 + R_2, R_3 \rightarrow -2R_1 + R_3$$

$$[A:B] \sim \begin{bmatrix} 1 & -2 & 3 & : & 2 \\ 0 & 5 & -5 & : & -2 \\ 0 & 5 & -8 & : & 1 \end{bmatrix}$$

$$R_3 \rightarrow -R_2 + R_3$$

$$[A:B] \sim \begin{bmatrix} 1 & -2 & 3 & : & 2 \\ 0 & 5 & -5 & : & -2 \\ 0 & 0 & -3 & : & 3 \end{bmatrix}$$

Hence we have, $x_1 - 2x_2 + 3x_3 = 2$
 $5x_2 - 5x_3 = -2$
 $-3x_3 = 3 \therefore x_3 = -1$

By back substitution, $5x_2 + 5 = -2 \therefore x_2 = -7/5 = -1.4$

Also, $x_1 - 2(-1.4) - 3 = 2 \therefore x_1 = 2.2$

Thus, $x_1 = 2.2, x_2 = -1.4, x_3 = -1$ is the required solution.

[19]. Apply Gauss elimination method to solve the system of equations.

$$2x + 5y + 7z = 52$$

$$2x + y - z = 0$$

$$x + y + z = 9$$

As it is convenient to have the leading coefficient as 1 we shall interchange the first and third equations. The augmented matrix will be,

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 2 & 1 & -1 & : & 0 \\ 2 & 5 & 7 & : & 52 \end{bmatrix}$$

It is convenient to have the leading coefficient as 1. Interchange the first and third equations. The augmented matrix will be

$$R_2 \rightarrow -2R_1 + R_2, R_3 \rightarrow -2R_1 + R_3$$

$$[A:B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -1 & -3 & : & -18 \\ 0 & 3 & 5 & : & 34 \end{bmatrix}$$

$$R_3 \rightarrow 3R_2 + R_3$$

$$[A:B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -1 & -3 & : & -18 \\ 0 & 0 & -4 & : & -20 \end{bmatrix}$$

Hence we have, $x + y + z = 9$

$$-y - 3z = -18$$

$$-4z = -20 \therefore z = 5$$

By back substitution, $y = 3$ and $x = 1$

Thus, $x = 1, y = 3, z = 5$ is the required solution.

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5.42 Gauss - Jordan elimination method

This method can be regarded as the modification of the previously discussed Gauss elimination method.

This method aims in reducing the coefficient matrix A to a diagonal matrix. This method is illustrated with reference to the system of equations (i) as in the article 5.41.

and How to solve these examples.
$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \right\} \text{--- (i)}$$

Step-1 This step is same as in Gauss elimination method.

Step-2 Referring to (iii) as in that method, we use the element $a_{22}' (\neq 0)$ to make the elements a_{12} and a_{32}' zero by elementary row transformations. With this transformation we obtain,

$$[A : B] \sim \left[\begin{array}{ccc|c} a_{11} & 0 & a_{13}' & b_1' \\ 0 & a_{22}' & a_{23}' & b_2' \\ 0 & 0 & a_{33}'' & b_3'' \end{array} \right]$$

$$[A : B] \sim \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22}' & a_{23}' & b_2' \\ 0 & a_{32}' & a_{33}' & b_3' \end{array} \right] \text{--- (ii)}$$

.... (v)

Step-3 Finally we use the element $a_{33}'' (\neq 0)$ to make the elements a_{23}' and a_{13}' zero by elementary row transformations. With this transformation we obtain,

$$[A : B] \sim \left[\begin{array}{ccc|c} a_{11} & 0 & 0 & b_1'' \\ 0 & a_{22}' & 0 & b_2'' \\ 0 & 0 & a_{33}'' & b_3'' \end{array} \right] \text{.... (vi)}$$

It can be easily seen from (vi) that,

$$a_{11}x_1 = b_1'', a_{22}'x_2 = b_2'', a_{33}''x_3 = b_3''$$

Thus we get the required x_1, x_2, x_3 being the exact solution.

WORKED PROBLEMS

[20] Solve the following system of equations by Gauss Jordan method.

$$x + y + z = 9$$

$$x - 2y + 3z = 8$$

$$2x + y - z = 3$$

☞ The augmented matrix of the system is

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 1 & -2 & 3 & : & 8 \\ 2 & 1 & -1 & : & 3 \end{bmatrix}$$

$$R_2 \rightarrow -R_1 + R_2, R_3 \rightarrow -2R_1 + R_3$$

$$[A : B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -3 & 2 & : & -1 \\ 0 & -1 & -3 & : & -15 \end{bmatrix}$$

We use the leading non zero entry in second row (-3) to make the element above 1 and below, 1 and -1 respectively zero.

$$R_1 \rightarrow R_2 + 3R_1, R_3 \rightarrow R_2 - 3R_3$$

$$[A : B] \sim \begin{bmatrix} 3 & 0 & 5 & : & 26 \\ 0 & -3 & 2 & : & -1 \\ 0 & 0 & 11 & : & 44 \end{bmatrix}$$

$$R_3 \rightarrow 1/11 \cdot R_3$$

$$\sim \begin{bmatrix} 3 & 0 & 5 & : & 26 \\ 0 & -3 & 2 & : & -1 \\ 0 & 0 & 1 & : & 4 \end{bmatrix}$$

(This step is performed only for convenience to get 1 as the leading entry in the third row).

We use the element 1 in the third row to make the two elements above 2 and 5 zero.

$$R_1 \rightarrow -5R_3 + R_1, R_2 \rightarrow -2R_3 + R_2$$

$$\sim \begin{bmatrix} 3 & 0 & 0 & : & 6 \\ 0 & -3 & 0 & : & -9 \\ 0 & 0 & 1 & : & 4 \end{bmatrix}$$

Hence, we have $3x = 6$, $-3y = -9$, $z = 4$

Thus, $\boxed{x = 2, y = 3, z = 4}$ is the required solution.

[21] Apply Gauss-Jordan method to solve the system of equations,

$$2x + 5y + 7z = 52$$

$$2x + y - z = 0$$

$$x + y + z = 9$$

As it is convenient to have the leading coefficient as 1 we shall interchange the first and third equations. The associated augmented matrix will be

$$[A : B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 2 & 1 & -1 & : & 0 \\ 2 & 5 & 7 & : & 52 \end{bmatrix}$$

$$R_2 \rightarrow -2R_1 + R_2, R_3 \rightarrow -2R_1 + R_3$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -1 & -3 & : & -18 \\ 0 & 3 & 5 & : & 34 \end{bmatrix}$$

$$R_1 \rightarrow R_2 + R_1, R_3 \rightarrow 3R_2 + R_3$$

$$\sim \begin{bmatrix} 1 & 0 & -2 & : & -9 \\ 0 & -1 & -3 & : & -18 \\ 0 & 0 & -4 & : & -20 \end{bmatrix}$$

$$R_3 \rightarrow -1/4 \cdot R_3$$

$$\sim \begin{bmatrix} 1 & 0 & -2 & : & -9 \\ 0 & -1 & -3 & : & -18 \\ 0 & 0 & 1 & : & 5 \end{bmatrix}$$

$$R_1 \rightarrow 2R_3 + R_1, R_2 \rightarrow 3R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & : & 1 \\ 0 & -1 & 0 & : & -3 \\ 0 & 0 & 1 & : & 5 \end{bmatrix}$$

$x \quad y \quad z : B(b_1, b_2, b_3)$

$$x, y, z : B = (b_1, b_2, b_3)$$

Hence, we have $x = 1, -y = -3, z = 5$,

Thus, $\boxed{x = 1, y = 3, z = 5}$ is the required solution.

[22] Apply Gauss - Jordan method to solve the following system of equations

$$2x_1 + x_2 + 3x_3 = 1$$

[June 2016]

$$4x_1 + 4x_2 + 7x_3 = 1$$

$$2x_1 + 5x_2 + 9x_3 = 3$$

The augmented matrix associated with the given system is,

$$[A : B] \sim \begin{bmatrix} 2 & 1 & 3 & : & 1 \\ 4 & 4 & 7 & : & 1 \\ 2 & 5 & 9 & : & 3 \end{bmatrix}$$

$$R_2 \rightarrow -2R_1 + R_2, R_3 \rightarrow -R_1 + R_3$$

$$\sim \begin{bmatrix} 2 & 1 & 3 & : & 1 \\ 0 & 2 & 1 & : & -1 \\ 0 & 4 & 6 & : & 2 \end{bmatrix}$$

$$R_1 \rightarrow -R_2 + 2R_1, R_3 \rightarrow -2R_2 + R_3$$

$$\sim \begin{bmatrix} 4 & 0 & 5 & : & 3 \\ 0 & 2 & 1 & : & -1 \\ 0 & 0 & 4 & : & 4 \end{bmatrix}$$

$$R_3 \rightarrow 1/4 \cdot R_3$$

$$(A:B) \sim \begin{bmatrix} 4 & 0 & 5 & : & 3 \\ 0 & 2 & 1 & : & -1 \\ 0 & 0 & 1 & : & 1 \end{bmatrix}$$

$$R_2 \rightarrow -R_3 + R_2, R_1 \rightarrow -5R_3 + R_1$$

$$\sim \begin{bmatrix} 4 & 0 & 0 & : & -2 \\ 0 & 2 & 0 & : & -2 \\ 0 & 0 & 1 & : & 1 \end{bmatrix}$$

Hence we have,

$$4x_1 = -2; 2x_2 = -2; x_3 = 1$$

Thus $x_1 = -0.5, x_2 = -1, x_3 = 1$ is the required solution.

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Remark: Iteration and approximation Both are same.

5.43 Gauss - Seidel method

Preamble : Iterative methods

The word 'Iterate' in the present context is to be understood as 'Do repeatedly'. The earlier discussed two methods for solving a system of linear equations are referred to as *direct methods* resulting in *exact solution* of the given system of equations. Iterative methods involve a *repetitive process* giving an *approximate solution*. In iterative methods we start with an approximate solution (initial approximation to the solution) and improve the accuracy of the same by repeating the prescribed procedure successively.

Computers and ultra modern calculators play a great role in iterative methods for obtaining the results to the desired degree of accuracy despite of data being large and complex.

Gauss - Seidel method is an iterative method for solving a system of linear algebraic equations which is *diagonally dominant* and the same is to be understood as follows.

Let us consider a system of three independent equations in three unknowns:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Unknowns



Diagonally dominant

This system of equations is said to be *diagonally dominant* if

$$|a_{11}| > |a_{12}| + |a_{13}|, |a_{22}| > |a_{21}| + |a_{23}|, |a_{33}| > |a_{31}| + |a_{32}|$$

Sometimes we may have to rearrange the given system of equations to meet this requirement. If this condition is satisfied, the solution exists as the iteration process will converge. *Gauss - Seidel iterative method* is illustrated for this system of equations.

We write the system of equations in the form

$$x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3] \quad \dots (1)$$

$$x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3] \quad \dots (2)$$

$$x_3 = \frac{1}{a_{33}} [b_3 - a_{31}x_1 - a_{32}x_2] \quad \dots (3)$$

We start with the trial solution (initial approximation)

$$x_1 = 0, x_2 = 0, x_3 = 0.$$

$x_1^{(0)} = x_1, x_2^{(0)} = x_2, x_3^{(0)} = x_3$ are initial approximation of x_1, x_2, x_3 variables.

MODULE - 5

The first approximations are as follows.

$$x_1^{(1)} = \frac{1}{a_{11}} [b_1 - a_{12} \cdot 0 - a_{13} \cdot 0] = \frac{b_1}{a_{11}}$$

This approximation is immediately used in (2) so that we have,

$$x_2^{(1)} = \frac{1}{a_{22}} [b_2 - a_{21} x_1^{(1)} - a_{23} \cdot 0]$$

$$\text{i.e., } x_2^{(1)} = \frac{1}{a_{22}} \left[b_2 - a_{21} \left(\frac{b_1}{a_{11}} \right) \right]$$

Finally, both these approximations are used in (3), so that we have

$$x_3^{(1)} = \frac{1}{a_{33}} [b_3 - a_{31} x_1^{(1)} - a_{32} x_2^{(1)}]$$

This completes first iteration.

The process is continued till we get the solution to the desired degree of accuracy.

WORKED PROBLEMS

[23] Solve the following system of equations by Gauss - Seidel method. [June 2016]

$$10x + y + z = 12$$

$$x + 10y + z = 12$$

$$x + y + 10z = 12$$

☞ The given system of equations are diagonally dominant and the equations are put in the following form.

$$x = \frac{1}{10} [12 - y - z] \quad \dots (1)$$

$$y = \frac{1}{10} [12 - x - z] \quad \dots (2)$$

$$z = \frac{1}{10} [12 - x - y] \quad \dots (3)$$

Let us start with the trial solution $x = 0, y = 0, z = 0$.

First iteration :

$$x^{(1)} = \frac{1}{10} [12 - 0 - 0] = 1.2$$

$$y^{(1)} = \frac{1}{10} [12 - 1.2 - 0] = 1.08$$

$$z^{(1)} = \frac{1}{10} [12 - 1.2 - 1.08] = 0.972$$

Second iteration :

$$x^{(2)} = \frac{1}{10} [12 - 1.08 - 0.972] = 0.9948$$

$$y^{(2)} = \frac{1}{10} [12 - 0.9948 - 0.972] = 1.00332$$

$$z^{(2)} = \frac{1}{10} [12 - 0.9948 - 1.00332] = 1.000188$$

Third iteration :

$$x^{(3)} = \frac{1}{10} [12 - 1.00332 - 1.000188] = 0.99965$$

$$y^{(3)} = \frac{1}{10} [12 - 0.99965 - 1.000188] = 1.00002$$

$$z^{(3)} = \frac{1}{10} [12 - 0.99965 - 1.00002] = 1.00003$$

Fourth iteration :

$$x^{(4)} = \frac{1}{10} [12 - 1.00002 - 1.00003] = 0.999995 \approx 1$$

$$y^{(4)} = \frac{1}{10} [12 - 1 - 1.00003] = 0.999997 \approx 1$$

$$z^{(4)} = \frac{1}{10} [12 - 1 - 1] = 1$$

Thus, $\boxed{x = 1, y = 1, z = 1}$

Remark : The iterative process could have been stopped at the third iteration itself as the solution can be approximated to $x = 1, y = 1, z = 1$.

[24] Solve the following system of equations by Gauss - Seidel method to obtain the final solution correct to three places of decimals

$$x + y + 54z = 110$$

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

☞ The given system of equations is not diagonally dominant and hence we have to first rearrange the given system of equations as follows.

$$27x + 6y - z = 85 \quad (|27| > |6| + |-1|)$$

$$6x + 15y + 2z = 72 \quad (|15| > |6| + |2|)$$

$$x + y + 54z = 110 \quad (|54| > |1| + |1|)$$

(Observe the coefficients column wise for rearrangement)

These equations are now written as follows.

$$x = \frac{1}{27} [85 - 6y + z] \quad \dots (1)$$

$$y = \frac{1}{15} [72 - 6x - 2z] \quad \dots (2)$$

$$z = \frac{1}{54} [110 - x - y] \quad \dots (3)$$

First iteration : We start with the trial solution $x = 0, y = 0, z = 0$.

$$x^{(1)} = \frac{1}{27} [85 - 0 + 0] = 3.14815$$

$$y^{(1)} = \frac{1}{15} [72 - 6(3.14815) - 0] = 3.54074$$

$$z^{(1)} = \frac{1}{54} [110 - 3.14815 - 3.54074] = 1.91317$$

Second iteration :

$$x^{(2)} = \frac{1}{27} [85 - 6(3.54074) + 1.91317] = 2.43218$$

$$y^{(2)} = \frac{1}{15} [72 - 6(2.43218) - 2(1.91317)] = 3.57204$$

$$z^{(2)} = \frac{1}{54} [110 - 2.43218 - 3.57204] = 1.92585 = 1.92585$$

Third iteration :

$$(10) \quad x^{(3)} = \frac{1}{27} [85 - 6(3.57204) + 1.92585] = 2.42569$$

$$(11) \quad y^{(3)} = \frac{1}{15} [72 - 6(2.42569) - 2(1.92585)] = 3.57294$$

$$(12) \quad z^{(3)} = \frac{1}{54} [110 - 2.42569 - 3.57294] = 1.92595$$

Fourth iteration :

$$(13) \quad x^{(4)} = \frac{1}{27} [85 - 6(3.57294) + 1.92595] = 2.42549$$

$$(14) \quad y^{(4)} = \frac{1}{15} [72 - 6(2.42549) - 2(1.92595)] = 3.57301$$

$$(15) \quad z^{(4)} = \frac{1}{54} [110 - 2.42549 - 3.57301] = 1.92595$$

It may be observed that the solution in the third and fourth iteration when approximated to three places of decimals is the same.

Thus, $x = 2.426, y = 3.573, z = 1.926$

[25] Solve the following system of equations by Gauss - Seidel method.

$$20x + y - 2z = 17$$

$$3x + 20y - z = -18$$

$$2x - 3y + 20z = 25$$

[Dec 2015, 16, 17, 18]

☞ The equations are diagonally dominant and hence we first write them in the following form.

$$x = \frac{1}{20} [17 - y + 2z]$$

$$y = \frac{1}{20} [-18 - 3x + z]$$

$$z = \frac{1}{20} [25 - 2x + 3y]$$

We start with the trial solution $x = 0, y = 0, z = 0$.

First iteration :

$$x^{(1)} = \frac{17}{20} = 0.85$$

$$y^{(1)} = \frac{1}{20} [-18 - 3(0.85)] = -1.0275$$

$$z^{(1)} = \frac{1}{20} [25 - 2(0.85) + 3(-1.0275)] = 1.0109$$

Second iteration :

$$x^{(2)} = \frac{1}{20} [17 - (-1.0275) + 2(1.0109)] = 1.0025$$

$$y^{(2)} = \frac{1}{20} [-18 - 3(1.0025) + 1.0109] = -0.9998$$

$$z^{(2)} = \frac{1}{20} [25 - 2(1.0025) + 3(-0.9998)] = 0.9998$$

Third iteration :

$$x^{(3)} = \frac{1}{20} [17 - (-0.9998) + 2(0.9998)] = 0.99997 \approx 1$$

$$y^{(3)} = \frac{1}{20} [-18 - 3(0.99997) + 0.9998] = -1.0000055 \approx -1$$

$$z^{(3)} = \frac{1}{20} [25 - 2(0.99997) + 3(-1.0000055)] = 1.0000022 \approx 1$$

Thus, $x = 1, y = -1, z = 1$ is the required solution.

[26] Employ Gauss - Seidel iteration method to solve.

$$5x + 2y + z = 12$$

$$x + 4y + 2z = 15$$

$$x + 2y + 5z = 20$$

Carryout 4 iterations taking the initial approximation to the solution as $(1, 0, 0)$

☞ The given system of equations are diagonally dominant and we put them in the following form.

$$x = \frac{1}{5} [12 - 2y - z]$$

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$$y = \frac{1}{4} [15 - x - 2z]$$

$$z = \frac{1}{5} [20 - x - 2y]$$

By data, $x^{(0)} = 1$, $y^{(0)} = 0$, $z^{(0)} = 3$

First iteration :

$$x^{(1)} = \frac{1}{5} [12 - 2(0) - 3] = 1.8$$

$$y^{(1)} = \frac{1}{4} [15 - 1.8 - 2(3)] = 1.8$$

$$z^{(1)} = \frac{1}{5} [20 - 1.8 - 2(1.8)] = 2.92$$

Second iteration :

$$x^{(2)} = \frac{1}{5} [12 - 2(1.8) - 2.92] = 1.096$$

$$y^{(2)} = \frac{1}{4} [15 - 1.096 - 2(2.92)] = 2.016$$

$$z^{(2)} = \frac{1}{5} [20 - 1.096 - 2(2.016)] = 2.9744$$

Third iteration :

$$x^{(3)} = \frac{1}{5} [12 - 2(2.016) - 2.9744] = 0.99872$$

$$y^{(3)} = \frac{1}{4} [15 - 0.99872 - 2(2.9744)] = 2.01312$$

$$z^{(3)} = \frac{1}{5} [20 - 0.99872 - 2(2.01312)] = 2.995$$

Fourth iteration :

$$x^{(4)} = \frac{1}{5} [12 - 2(2.01312) - 2.995] = 0.995752$$

$$y^{(4)} = \frac{1}{4} [15 - 0.995752 - 2(2.995)] = 2.003562$$

$$z^{(4)} = \frac{1}{5} [20 - 0.995752 - 2(2.003562)] = 2.9994248$$

Thus the solution after four iterations correct to four decimal places is given by

$x = 0.9958, y = 2.0036, z = 2.9994.$

ASSIGNMENT

Solve the following system of equations by Gauss elimination method and also by Gauss-Jordan method.

1. $x + 2y + z = 3, 2x + 3y + 3z = 10, 3x - y + 2z = 13$

2. $2x + 3y - z = 5, 4x + 4y - 3z = 3, 2x - 3y + 2z = 2$

3. $3x + 4y + 5z = 18, 2x - y + 8z = 13, 5x - 2y + 7z = 20$

4. Solve the following system of equations by Gauss-Seidel method. Perform three iterations and present the solution correct to three decimal places.

$$28x + 4y - z = 32, 2x + 17y + 4z = 35, x + 3y + 10z = 24$$

ANSWERS

Note: Answers are given in the respective order of the variables as in the given equations

1. 2, -1, 3

2. 1, 2, 3

3. 3, 1, 1

4. 0.994, 1.507, 1.849

5.5 Eigen values and Eigen vectors of a square matrix

Definition : Given a square matrix A , if there exists a scalar λ (real or complex) and a non zero column matrix X such that $AX = \lambda X$, then λ is called an *eigen value* of A and X is called an *eigen vector* of A corresponding to an eigen value λ .

If I is the unit matrix of the same order as that of A , we have $X = IX$ and hence $AX = \lambda X$ can be written as

$$AX = \lambda (IX) = (\lambda I) X$$

ie., $[A - \lambda I] [X] = [0]$, $[0]$ is the null matrix.

Let us consider a square matrix of order 3 represented by

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}, \text{ Also } \lambda I = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\therefore [A - \lambda I] = \begin{bmatrix} (a_1 - \lambda) & a_2 & a_3 \\ b_1 & (b_2 - \lambda) & b_3 \\ c_1 & c_2 & (c_3 - \lambda) \end{bmatrix}; \text{ Also let } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

It can be easily seen that $[A - \lambda I][X] = [0]$ represents a set of homogenous equations (RHS being zero) in 3 unknowns.

$$\text{ie., } (a_1 - \lambda)x + a_2y + a_3z = 0$$

$$b_1x + (b_2 - \lambda)y + b_3z = 0$$

$$c_1x + c_2y + (c_3 - \lambda)z = 0$$

A nontrivial solution (atleast one of $x, y, z \neq 0$) for this system exists if the determinant of the coefficient matrix is zero.

$$\text{ie., } \begin{vmatrix} (a_1 - \lambda) & a_2 & a_3 \\ b_1 & (b_2 - \lambda) & b_3 \\ c_1 & c_2 & (c_3 - \lambda) \end{vmatrix} = 0$$

On expanding, we get a cubic equation in λ which is called the *characteristic equation* of A . The roots of this equation are the *eigen values* which are also called *eigen roots* or *characteristic roots* or *latent roots*. For each value of λ there will be an eigen vector $X \neq 0$ which is also called a *characteristic vector*.

Note : Computation of eigen values and eigen vectors is restricted to a second order square matrix. It is necessary in the discussion of Diagonalisation of a square matrix of order 2 presented in the article 5.7.

An illustrative example for a third order square matrix

Find all the eigen values and the corresponding eigen vectors of the matrix.

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Soln. The characteristic equation of A is $|A - \lambda I| = 0$.

$$\text{ie., } \begin{vmatrix} (8 - \lambda) & -6 & 2 \\ -6 & (7 - \lambda) & -4 \\ 2 & -4 & (3 - \lambda) \end{vmatrix} = 0$$

On expanding we have,

$$(8 - \lambda) [(7 - \lambda)(3 - \lambda) - 16] + 6 [-6(3 - \lambda) + 8] + 2 [24 - 2(7 - \lambda)] = 0$$

$$\text{ie., } (8 - \lambda) [5 - 10\lambda + \lambda^2] + 6 [6\lambda - 10] + 2 [10 + 2\lambda] = 0$$

$$\text{ie., } -\lambda^3 + 18\lambda^2 - 45\lambda = 0, \text{ on simplification.}$$

$$\text{or } \lambda^3 - 18\lambda^2 + 45\lambda = 0$$

$$\begin{aligned} 24 - 7\lambda - 3\lambda + 2\lambda - 16 &= 5 - 10\lambda + \lambda^2 \\ -18 + 6\lambda + 8 &= -10 + 2\lambda \\ \lambda^2 - 10\lambda + 2\lambda &= 10 - 2\lambda \end{aligned}$$

$$\text{ie., } \lambda (\lambda^2 - 18\lambda + 45) = 0 \text{ or } \lambda (\lambda - 3) (\lambda - 15) = 0$$

$\therefore \lambda = 0, 3, 15$ are the eigen values of A.

We now form the system of equations

$$\left. \begin{aligned} (8 - \lambda)x - 6y + 2z &= 0 \\ -6x + (7 - \lambda)y - 4z &= 0 \\ 2x - 4y + (3 - \lambda)z &= 0 \end{aligned} \right\} \dots (1)$$

Case - i : Let $\lambda = 0$. The system of equations become

$$8x - 6y + 2z = 0 \dots (i)$$

$$-6x + 7y - 4z = 0 \dots (ii)$$

$$2x - 4y + 3z = 0 \dots (iii)$$

Applying the rule of cross multiplication for (i) and (ii),

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ 7 & -4 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 8 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}}$$

$$\text{ie., } \frac{x}{10} = \frac{y}{20} = \frac{z}{20} \text{ or } \frac{x}{1} = \frac{y}{2} = \frac{z}{2}$$

$\therefore (x, y, z)$ are proportional to $(1, 2, 2)$ and we can write $x = k, y = 2k, z = 2k$ where k is arbitrary. However it is enough to keep the values of (x, y, z) in the simplest form $x = 1, y = 2, z = 2$. These values satisfy all the equations simultaneously.

Eigen vectors

Thus the eigen vector X_1 corresponding to the eigen value $\lambda = 0$ is $X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

[The same will be written as a row vector in the form $X_1 = (1, 2, 2)$ in future]

$X_1 = (1, 2, 2)$ is the eigen vector corresponding to $\lambda = 0$.

Case - ii : Let $\lambda = 3$ and the corresponding equations from (1) are

$$5x - 6y + 2z = 0 \dots (iv)$$

$$-6x + 4y - 4z = 0 \dots (v)$$

$$2x - 4y + 0z = 0 \dots (vi)$$

From (iv) and (v) we have as before,

$$\frac{x}{24 - 8} = \frac{-y}{-20 + 12} = \frac{z}{20 - 36} \text{ or } \frac{x}{16} = \frac{-y}{-8} = \frac{z}{-16}$$

ie., $\frac{x}{2} = \frac{y}{1} = \frac{z}{-2} \therefore (x, y, z) = (2, 1, -2)$

$X_2 = (2, 1, -2)$ is the eigen vector corresponding to $\lambda = 3$.

Case - iii : Let $\lambda = 15$ and the associated equations from (1) are

$$-7x - 6y + 2z = 0 \quad \dots \text{(vii)}$$

$$-6x - 8y - 4z = 0 \quad \dots \text{(viii)}$$

$$2x - 4y - 12z = 0 \quad \dots \text{(ix)}$$

From (vii) and (viii) we have,

$$\frac{x}{24+16} = \frac{-y}{28+12} = \frac{z}{56-36} \quad \text{or} \quad \frac{x}{40} = \frac{-y}{40} = \frac{z}{20}$$

ie., $\frac{x}{2} = \frac{y}{-2} = \frac{z}{1} \therefore (x, y, z) = (2, -2, 1)$

$X_3 = (2, -2, 1)$ is the eigen vector corresponding to $\lambda = 15$.

5.6 Rayleigh's power method

This is an iterative method to determine the numerically largest eigen value (dominant eigen value) and the corresponding eigen vector of a square matrix.

Flow chart for solving problems

Suppose A is the given square matrix, we assume initially an eigen vector (column matrix) X_0 in a simple form like $[1, 0, 0]'$ or $[0, 1, 0]'$ or $[0, 0, 1]'$ or $[1, 1, 1]'$ and find the matrix product AX_0 which will also be a column matrix.

We take out the largest element as the common factor (this technique is called normalization) to obtain $AX_0 = \lambda^{(1)} X^{(1)}$.

We then compute $AX^{(1)}$ and again put it in the form $AX^{(1)} = \lambda^{(2)} X^{(2)}$ by normalization.

This iterative process is continued till two consecutive iterative values of λ and X are same upto a desired degree of accuracy.

The values so obtained are respectively the largest eigen value (λ) and the corresponding eigen vector (X) of the given square matrix A .

WORKED PROBLEMS

[27] Find the largest eigen value and the corresponding eigen vector of the matrix A by the power method given that

[June 2016, 18, Dec. 16, 18]

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Here

$\lambda \rightarrow$ eigen value
 $X \rightarrow$ eigen vectors

Let $X^{(0)} = [1, 0, 0]'$ be the initial eigen vector.

$$\therefore AX^{(0)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \lambda^{(1)} X^{(1)}$$

$$AX^{(1)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 0 \\ 2 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \lambda^{(2)} X^{(2)}$$

$$AX^{(2)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 2.8 \\ 0 \\ 2.6 \end{bmatrix} = 2.8 \begin{bmatrix} 1 \\ 0 \\ 0.93 \end{bmatrix} = \lambda^{(3)} X^{(3)}$$

$$AX^{(3)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.93 \end{bmatrix} = \begin{bmatrix} 2.93 \\ 0 \\ 2.86 \end{bmatrix} = 2.93 \begin{bmatrix} 1 \\ 0 \\ 0.98 \end{bmatrix} = \lambda^{(4)} X^{(4)}$$

$$AX^{(4)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.98 \end{bmatrix} = \begin{bmatrix} 2.98 \\ 0 \\ 2.96 \end{bmatrix} = 2.98 \begin{bmatrix} 1 \\ 0 \\ 0.99 \end{bmatrix} = \lambda^{(5)} X^{(5)}$$

$$AX^{(5)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.99 \end{bmatrix} = \begin{bmatrix} 2.99 \\ 0 \\ 2.98 \end{bmatrix} = 2.99 \begin{bmatrix} 1 \\ 0 \\ 0.997 \end{bmatrix} = \lambda^{(6)} X^{(6)}$$

$$AX^{(6)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.997 \end{bmatrix} = \begin{bmatrix} 2.997 \\ 0 \\ 2.994 \end{bmatrix} = 2.997 \begin{bmatrix} 1 \\ 0 \\ 0.999 \end{bmatrix} = \lambda^{(7)} X^{(7)}$$

Thus the **largest eigen value** is approximately 3 and the corresponding **eigen vector** is $[1, 0, 1]'$

[28] Find the dominant eigen value and the corresponding eigen vector of the matrix

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

[June, Dec. 2017]

by power method taking the initial eigen vector as $[1, 1, 1]'$.

☞ By data $X^{(0)} = [1, 1, 1]'$ is the initial eigen vector.

$$AX^{(0)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 4 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 0 \\ 0.67 \end{bmatrix} = \lambda^{(1)} X^{(1)}$$

$$AX^{(1)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.67 \end{bmatrix} = \begin{bmatrix} 7.34 \\ -2.67 \\ 4.01 \end{bmatrix} = 7.34 \begin{bmatrix} 1 \\ -0.36 \\ 0.55 \end{bmatrix} = \lambda^{(2)} X^{(2)}$$

$$AX^{(2)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.36 \\ 0.55 \end{bmatrix} = \begin{bmatrix} 7.82 \\ -3.63 \\ 4.01 \end{bmatrix} = 7.82 \begin{bmatrix} 1 \\ -0.46 \\ 0.51 \end{bmatrix} = \lambda^{(3)} X^{(3)}$$

$$AX^{(3)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.46 \\ 0.51 \end{bmatrix} = \begin{bmatrix} 7.94 \\ -3.89 \\ 3.99 \end{bmatrix} = 7.94 \begin{bmatrix} 1 \\ -0.49 \\ 0.5 \end{bmatrix} = \lambda^{(4)} X^{(4)}$$

$$AX^{(4)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.49 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 7.98 \\ -3.97 \\ 3.99 \end{bmatrix} = 7.98 \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \lambda^{(5)} X^{(5)}$$

$$AX^{(5)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 8 \\ -4 \\ 4 \end{bmatrix} = 8 \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \lambda^{(6)} X^{(6)}$$

Thus the dominant eigen value is 8 and the corresponding eigen vector is $[1, -0.5, 0.5]'$ or $[2, -1, 1]'$ equivalently.

[29] Using Rayleigh's power method find numerically the largest eigen value and the corresponding eigen vector of the matrix.

$$A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$$

☞ Let $X^{(0)} = [1, 0, 0]'$ be the initial eigen vector.

$$AX^{(0)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 25 \\ 1 \\ 2 \end{bmatrix} = 25 \begin{bmatrix} 1 \\ 0.04 \\ 0.08 \end{bmatrix} = \lambda^{(1)} X^{(1)}$$

$$AX^{(1)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0.04 \\ 0.08 \end{bmatrix} = \begin{bmatrix} 25.2 \\ 1.12 \\ 1.68 \end{bmatrix} = 25.2 \begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix} = \lambda^{(2)} X^{(2)}$$

$$AX^{(2)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix} = \begin{bmatrix} 25.18 \\ 1.12 \\ 1.72 \end{bmatrix} = 25.18 \begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix} = \lambda^{(3)} X^{(3)}$$

We observe that $X^{(2)} = X^{(3)}$

Thus the numerically largest eigen value of A is **25.18** and the corresponding eigen vector is $[1, 0.04, 0.07]^T$

[30] Find the numerically largest eigen value and the corresponding eigen vector of

the matrix $A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ by taking the initial approximation to the eigen

vector as $[1, 0.8, -0.8]^T$. Perform 5 iterations.

☞ By data $X^{(0)} = [1, 0.8, -0.8]^T$ is the initial eigen vector. Hence we have,

$$AX^{(0)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.8 \\ -0.8 \end{bmatrix} = \begin{bmatrix} 5.6 \\ 5.2 \\ -5.2 \end{bmatrix} = 5.6 \begin{bmatrix} 1 \\ 0.93 \\ -0.93 \end{bmatrix} = \lambda^{(1)} X^{(1)}$$

$$AX^{(1)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.93 \\ -0.93 \end{bmatrix} = \begin{bmatrix} 5.86 \\ 5.72 \\ -5.72 \end{bmatrix} = 5.86 \begin{bmatrix} 1 \\ 0.98 \\ -0.98 \end{bmatrix} = \lambda^{(2)} X^{(2)}$$

$$AX^{(2)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.98 \\ -0.98 \end{bmatrix} = \begin{bmatrix} 5.96 \\ 5.92 \\ -5.92 \end{bmatrix} = 5.96 \begin{bmatrix} 1 \\ 0.99 \\ -0.99 \end{bmatrix} = \lambda^{(3)} X^{(3)}$$

$$AX^{(3)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.99 \\ -0.99 \end{bmatrix} = \begin{bmatrix} 5.98 \\ 5.96 \\ -5.96 \end{bmatrix} = 5.98 \begin{bmatrix} 1 \\ 0.997 \\ -0.997 \end{bmatrix} = \lambda^{(4)} X^{(4)}$$

$$AX^{(4)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.997 \\ -0.997 \end{bmatrix} = \begin{bmatrix} 5.994 \\ 5.988 \\ -5.988 \end{bmatrix} = 5.994 \begin{bmatrix} 1 \\ 0.999 \\ -0.999 \end{bmatrix} = \lambda^{(5)} X^{(5)}$$

Thus after 5 iterations the numerically largest eigen value is **5.994** and the corresponding eigen vector is $[1, 0.999, -0.999]^T$

[31] Find the largest eigen value and the corresponding eigen vector of the matrix A , by using the power method by taking initial vector as $[1, 1, 1]^T$

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

[Dec 2015, June 18]

☞ By data $X^{(0)} = [1, 1, 1]^T$. Hence we have

$$AX^{(0)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \lambda^{(1)} X^{(1)}$$

$$AX^{(1)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \lambda^{(2)} X^{(2)}$$

$$AX^{(2)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \\ 3 \end{bmatrix} = 4 \begin{bmatrix} 0.75 \\ -1 \\ 0.75 \end{bmatrix} = \lambda^{(3)} X^{(3)}$$

$$AX^{(3)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.75 \\ -1 \\ 0.75 \end{bmatrix} = \begin{bmatrix} 2.5 \\ -3.5 \\ 2.5 \end{bmatrix} = 3.5 \begin{bmatrix} 0.71 \\ -1 \\ 0.71 \end{bmatrix} = \lambda^{(4)} X^{(4)}$$

$$AX^{(4)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.71 \\ -1 \\ 0.71 \end{bmatrix} = \begin{bmatrix} 2.42 \\ -3.42 \\ 2.42 \end{bmatrix} = 3.42 \begin{bmatrix} 0.708 \\ -1 \\ 0.708 \end{bmatrix} = \lambda^{(5)} X^{(5)}$$

$$AX^{(5)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.708 \\ -1 \\ 0.708 \end{bmatrix} = \begin{bmatrix} 2.416 \\ -3.416 \\ 2.416 \end{bmatrix} = 3.416 \begin{bmatrix} 0.7073 \\ -1 \\ 0.7073 \end{bmatrix} = \lambda^{(6)} X^{(6)}$$

$$AX^{(6)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.7073 \\ -1 \\ 0.7073 \end{bmatrix} = \begin{bmatrix} 2.4146 \\ -3.4146 \\ 2.4146 \end{bmatrix} = 3.4146 \begin{bmatrix} 0.7071 \\ -1 \\ 0.7071 \end{bmatrix} = \lambda^{(7)} X^{(7)}$$

Hence the largest eigen value is 3.4146 and the corresponding eigen vector is $[0.7071, -1, 0.7071]'$

ASSIGNMENT

Use Rayleigh's power method to find the largest eigen value and the corresponding eigen vector of the following matrices.

1. $\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ (Take $X_0 = [1, 0, 0]'$)

2. $\begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix}$ (Take $X_0 = [0, 1, 0]'$)

3. $\begin{bmatrix} 10 & 2 & 1 \\ 2 & 10 & 1 \\ 2 & 1 & 10 \end{bmatrix}$ (Take $X_0 = [0, 0, 1]'$)

ANSWERS

1. $\lambda = 4; [1, 0.5, 0]'$ 2. $\lambda = 6; [1, 0, 1]'$ 3. $\lambda = 13; [1, 1, 1]'$

$$\text{ie., } \begin{bmatrix} (-1-\lambda) & 3 \\ -2 & (4-\lambda) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{ie., } \begin{cases} (-1-\lambda)x + 3y = 0 \\ -2x + (4-\lambda)y = 0 \end{cases}$$

Case - (i) : Let $\lambda = 1$.

$$\text{We get } -2x + 3y = 0 \text{ or } 2x = 3y \text{ or } \frac{x}{3} = \frac{y}{2}$$

$\therefore X_1 = (3, 2)'$ is the eigen vector corresponding to $\lambda = 1$.

Case - (ii) : Let $\lambda = 2$

$$\text{We get } -3x + 3y = 0 \text{ or } x = y \text{ or } \frac{x}{1} = \frac{y}{1}$$

$\therefore X_2 = (1, 1)'$ is the eigen vector corresponding to $\lambda = 2$.

$$\text{Modal matrix } P = [X_1 X_2] = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\text{We have } |P| = 1 \text{ and } P^{-1} = \frac{1}{|P|} (\text{Adj } P)$$

$$P^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$\text{Now } P^{-1}AP = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\therefore P^{-1}AP = D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \text{ is the diagonal matrix.}$$

Thus,

$$P^{-1}AP = \text{Diag } (1, 2)$$

[33] Diagonalize the matrix $\begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$

[June 2016, 17, 18]

Let $A = \begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$

The characteristic equation of A is $|A - \lambda I| = 0$.

$$\text{ie., } \begin{vmatrix} (-19 - \lambda) & 7 \\ -42 & (16 - \lambda) \end{vmatrix} = 0$$

$$\text{ie., } \lambda^2 + 3\lambda - 304 + 294 = 0$$

$$\text{ie., } \lambda^2 + 3\lambda - 10 = 0$$

$$\text{or } (\lambda - 2)(\lambda + 5) = 0$$

$$\text{ie., } \lambda = 2, -5 \text{ are the eigen values of } A.$$

$$\text{Now consider, } [A - \lambda I][X] = [0]$$

$$\text{ie., } \begin{bmatrix} (-19 - \lambda) & 7 \\ -42 & (16 - \lambda) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{ie., } (-19 - \lambda)x + 7y = 0$$

$$-42x + (16 - \lambda)y = 0$$

Case - (i) : Let $\lambda = 2$

$$\text{We get } -21x + 7y = 0 \text{ and } -42x + 14y = 0$$

$$\text{ie., } y = 3x \text{ or } \frac{y}{3} = \frac{x}{1}$$

$\therefore X_1 = (1, 3)'$ is the eigen vector corresponding to $\lambda = 2$.

Case - (ii) : Let $\lambda = -5$

$$\text{We get } -14x + 7y = 0 \text{ and } -42x + 21y = 0$$

$$\text{ie., } y = 2x \text{ or } \frac{y}{2} = \frac{x}{1}$$

$\therefore X_2 = (1, 2)'$ is the eigen vector corresponding to $\lambda = -5$.

Modal matrix $P = [X_1 X_2] = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$

We have, $|P| = 2 - 3 = -1$ and $P^{-1} = \frac{1}{|P|} (\text{Adj } P)$

$$P^{-1} = - \begin{bmatrix} 2 & -1 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix}$$

Now $P^{-1}AP = D = \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} -4 & 2 \\ -15 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$

$\therefore P^{-1}AP = D = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$ is the diagonal matrix.

Thus,

$$P^{-1}AP = \text{Diag } (2, -5)$$

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